

Cloud Repatriation

Bring Your VMs Home with h2kvm + hypersdk

Export VMs from AWS, Azure, GCP, Oracle Cloud, Alibaba Cloud, and OpenStack. Convert to QCOW2 with full guest fixes. Deploy to on-premises KVM or KubeVirt. 10 cloud providers, one tool, zero cloud lock-in. Save 70-90% on compute costs by running workloads on infrastructure you own.

10 Cloud Providers | \$0 Migration Tool | 70-90% Cost Savings | Proprietary (Zyvor AI Labs)

Why Repatriate?

The cloud promise was cost savings and agility. The reality is vendor lock-in and runaway bills.

\$3.6M

Average 3-year overspend
on cloud vs on-prem for 50 VMs

72%

Of enterprises report
cloud costs exceeded budget

\$0.09

Per GB egress on AWS/Azure
— the hidden tax on your data

94%

Of organizations use
multi-cloud (complexity cost)

Runaway Costs

- Cloud bills grow 20-30% annually
- Reserved instances require 1-3 year commitment
- Spot/preemptible instances get terminated
- Egress fees on every data transfer
- Hidden costs: NAT gateways, load balancers, DNS queries
- **On-prem KVM: fixed, predictable, amortized**

Data Sovereignty

- GDPR requires data stays in specific jurisdictions
- HIPAA: patient data on infrastructure you control
- Government: classified data cannot leave the building
- Financial: PCI-DSS and SOX audit requirements
- Cloud regions may not match your compliance needs
- **On-prem: data never leaves your facility**

Vendor Lock-In

- AWS AMIs don't run on Azure or GCP
- Managed services (RDS, DynamoDB) are proprietary
- Cloud-specific IAM, networking, storage APIs
- Export tools are intentionally limited
- Cloud vendors don't want you to leave
- **KVM + QCOW2: open formats, no lock-in**

Cloud (Before Repatriation)

- \$8,000-15,000/month per workload cluster
- Egress fees on every API call and backup

On-Prem KVM (After Repatriation)

- \$500-2,000/month (hardware amortized over 5 years)
- Zero egress fees — data stays local

- Locked to one provider's formats and APIs
- Unpredictable monthly bills (usage spikes)
- Compliance risk (data location uncertainty)
- Dependency on provider uptime and pricing

- Open formats: QCOW2, libvirt XML, KubeVirt YAML
- Fixed, predictable costs — no surprises
- Full data sovereignty — physical control
- You own the infrastructure, no vendor dependency

10 Cloud Providers

hypersdk exports VMs from every major cloud. One unified API. One tool to learn.

Cloud Provider	Export Method	Auth	Disk Format	Guest Fixes Applied	Target
AWS EC2	EBS snapshot → S3 export → download	IAM / access keys	VMDK or RAW	Remove SSM/EC2Config, fix fstab (xvd→UUID), rebuild initramfs	KVM / KubeVirt
Azure VMs	Managed disk snapshot → SAS URL → download	Service principal	VHD	Remove waagent + Hyper-V IC, fix fstab, rebuild initramfs	KVM / KubeVirt
GCP Compute	Persistent disk → GCS image export → download	Service account	RAW (tar.gz)	Remove guest-agent, fix fstab, rebuild initramfs, disable cloud-init	KVM / KubeVirt
OCI	Boot volume → Object Storage → download	API key	QCOW2 or RAW	Remove OCI agent, fix fstab, rebuild initramfs	KVM / KubeVirt
Alibaba ECS	ECS disk snapshot → OSS → download	AccessKey	RAW or VHD	Remove Alibaba agent, fix fstab, rebuild initramfs	KVM / KubeVirt
OpenStack	Glance image or Cinder volume export	Keystone v3	QCOW2 or RAW	Disable cloud-init, fix fstab, rebuild initramfs	KVM / KubeVirt
Proxmox VE	API-based disk export	Token / PAM	QCOW2 or RAW	Standard KVM fixes (minimal — already KVM-based)	KVM / KubeVirt
KubeVirt	PVC / DataVolume export	kubeconfig	RAW	Minimal — already KVM-compatible	KVM / KubeVirt
VMware vSphere	govc OVA/OVF, HTTPS	SSO / credentials	VMDK	Remove VMware Tools, fix fstab, rebuild initramfs, VirtIO	KVM / KubeVirt
Hyper-V	VHD/VHDX direct copy or WinRM	WinRM / SSH	VHD / VHDX	Remove Hyper-V IC, fix fstab, rebuild initramfs, VirtIO	KVM / KubeVirt

No other migration tool supports all 10. AWS MGN only exports from AWS. Azure Migrate only from Azure. virt-v2v handles zero cloud sources. Red Hat MTV handles VMware only. h2kvm + hypersdk is the *only* unified cloud repatriation platform.

How It Works

The complete repatriation pipeline — from cloud VM to running on-prem.

AWS EC2 Repatriation



Azure VM Repatriation



GCP Compute Repatriation



Step-by-Step Pipeline

#	Step	Component	What Happens	Output
1	Authenticate	hypersdk	Connect to cloud provider API (IAM, service principal, API key)	Authenticated session
2	Discover VMs	hypersdk	List VMs with CPU, RAM, disk, OS, tags — filter by name, tag, region	VM inventory with metadata
3	Snapshot	hypersdk	Create consistent disk snapshot (optional: stop VM first for DBs)	Point-in-time disk snapshot
4	Export	hypersdk	Export disk image to object storage (S3/Blob/GCS), download locally	VMDK, VHD, or RAW image file
5	Convert	h2kvm	Convert to QCOW2, flatten, optional compression	Optimized QCOW2 image
6	Guest Fix	h2kvm	fstab UUID repair, initramfs rebuild (VirtIO), GRUB fix, remove cloud agents, SELinux relabel, Windows VirtIO injection	Boot-ready for KVM

7	Deploy	h2kvm	Generate libvirt XML or KubeVirt manifest, virsh define or kubectl apply	VM running on-prem
8	Verify	h2kvm	Automated boot test, service health check, network validation	Confirmed working
9	Notify	hypersdk	Webhook to Slack/Teams/PagerDuty with pass/fail per VM	Team informed
10	Decommission	Manual	Terminate cloud VM after validation period (keep snapshot as backup)	Cloud bill drops to \$0

Cost Savings by Provider

Real numbers: what you spend in the cloud vs what you'd spend on-prem with KVM.

AWS EC2 Repatriation — 20 VMs, 3-Year TCO

Cost Item	AWS EC2	On-Prem KVM	Savings
Compute (m5.2xlarge x 20)	\$528,000	\$40,000 (hardware)	\$488,000
EBS Storage (20TB)	\$72,000	\$8,000 (NVMe SSDs)	\$64,000
Data Transfer / Egress	\$97,200	\$0	\$97,200
NAT Gateway	\$31,536	\$0	\$31,536
Load Balancers	\$15,768	\$0 (HAProxy/Nginx)	\$15,768
Migration (one-time)	N/A	\$5,000	-\$5,000
Operations	\$54,000	\$45,000	\$9,000
3-Year Total	\$798,504	\$98,000	\$700,504

Azure Repatriation — 20 VMs, 3-Year TCO

Cost Item	Azure	On-Prem KVM	Savings
Compute (D8s_v3 x 20)	\$460,800	\$40,000	\$420,800
Managed Disks (20TB)	\$88,560	\$8,000	\$80,560
Bandwidth / Egress	\$77,760	\$0	\$77,760
Azure Monitor + Log Analytics	\$21,600	\$0 (Prometheus)	\$21,600
Migration	N/A	\$5,000	-\$5,000
Operations	\$54,000	\$45,000	\$9,000
3-Year Total	\$702,720	\$98,000	\$604,720

GCP Repatriation — 20 VMs, 3-Year TCO

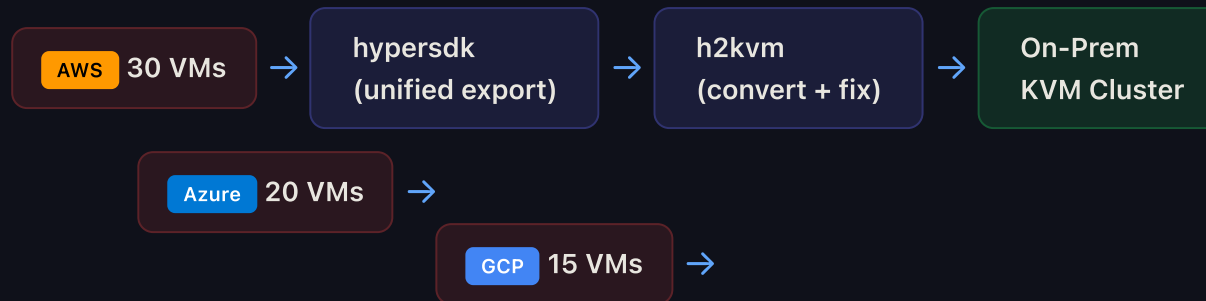
Cost Item	GCP	On-Prem KVM	Savings
Compute (n2-standard-8 x 20)	\$403,200	\$40,000	\$363,200
Persistent Disk (20TB)	\$58,320	\$8,000	\$50,320
Egress	\$69,984	\$0	\$69,984
Cloud Logging + Monitoring	\$18,000	\$0 (Prometheus/Grafana)	\$18,000
Migration	N/A	\$5,000	-\$5,000
Operations	\$54,000	\$45,000	\$9,000
3-Year Total	\$603,504	\$98,000	\$505,504

Average 3-Year Savings: \$600K per 20 VMs

AWS: \$700K | Azure: \$605K | GCP: \$506K
Scale linearly — 100 VMs = \$2.5M-\$3.5M saved over 3 years.

Multi-Cloud Consolidation

VMs scattered across AWS, Azure, and GCP? Consolidate to one on-prem KVM cluster.



The Multi-Cloud Problem

Before: Multi-Cloud Sprawl

- 3 cloud consoles, 3 billing systems, 3 IAM models
- Each cloud has different VM formats, networking, storage
- 3x the monitoring tools (CloudWatch + Azure Monitor + Cloud Logging)
- 3x the security audits, 3x the compliance reviews
- Staff need expertise in all 3 platforms
- Cross-cloud data transfer fees compound costs
- **Total: \$180K+/yr in cloud bills + \$200K+/yr in multi-cloud ops**

After: Consolidated On-Prem

- One platform: KVM/libvirt or K3s + KubeVirt
- One format: QCOW2 + libvirt XML (open standards)
- One monitoring: Prometheus + Grafana (free)
- One security audit, one compliance framework
- Staff learn one platform, not three
- Zero cross-cloud and egress fees
- **Total: \$30K/yr hardware + \$80K/yr in ops = \$110K/yr**

hypersdk Unified Provider Interface

```
# Same API for every cloud – just change the provider config:
```

```
# AWS export
```

```
hypersdk export --provider aws --vm-id i-0abc123def456 --output ./exports/
```

```
# Azure export
```

```
hypersdk export --provider azure --vm-id /subscriptions/.../vm-name --output ./exports/
```

```
# GCP export
```

```
hypersdk export --provider gcp --vm-id projects/my-proj/zones/us-central1-a/instances/vm-1 --output ./exports/
```

```
# Then convert ALL of them with onetool:  
h2kvmctl --config consolidate.yaml # Processes all exports, applies per-source fixes
```

Cloud-Specific Technical Details

What h2kvm fixes for each cloud provider — Linux and Windows.

AWS EC2 Specifics

- **Export:** EBS snapshot → CreateStoreImageTask or `ec2 create-snapshot + export-image` to S3
- **Linux fixes:** Remove AWS SSM agent, EC2 instance connect, CloudWatch agent. Fix fstab from `/dev/xvda` to UUID. Rebuild initramfs: replace `xen-blkfront` with `virtio_blk`
- **Windows fixes:** Remove EC2Config/EC2Launch, SSM agent, CloudWatch. Inject VirtIO drivers, fix BCD
- **Networking:** Remove ENI-specific configs, elastic IP associations. Clean `/etc/sysconfig` or `netplan`
- **cloud-init:** Disable EC2 datasource, preserve hostname and SSH keys

Azure VM Specifics

- **Export:** Managed disk snapshot → grant access (SAS URL) → download VHD via `azcopy` or `wget`
- **Linux fixes:** Remove `walinuxagent` (`waagent`), Hyper-V IC daemons (`hv_kvp`, `hv_vss`). Fix fstab from `/dev/sda` to UUID. Rebuild initramfs: replace `hv_storvsc` with `virtio_blk`
- **Windows fixes:** Remove Azure VM agent, Hyper-V IC. Inject VirtIO, fix BCD. Remove Azure diagnostics extension
- **Networking:** Remove Azure-specific accelerated networking config. Clean udev rules for Mellanox/`hv_netvsc`
- **cloud-init:** Disable Azure datasource, remove CustomData provisioning

GCP Compute Specifics

- **Export:** `gcloud compute images export` → GCS bucket → download `tar.gz` containing `disk.raw`
- **Linux fixes:** Remove `google-guest-agent`, `google-osconfig-agent`, `google-compute-engine` packages. Fix fstab from `/dev/sda` to UUID. Rebuild initramfs with VirtIO
- **Windows fixes:** Remove GCP compute agent, metadata scripts. Inject VirtIO, fix BCD
- **Networking:** Remove `google-guest-agent` NIC management. Clean `/etc/netplan` or `ifcfg` with GCP metadata references
- **cloud-init:** Disable GCE datasource, remove startup-script hooks

OCI / Alibaba / OpenStack

- **OCI:** Boot volume → Object Storage pre-authenticated request → download. Remove `oracle-cloud-agent`, fix fstab, rebuild initramfs
- **Alibaba:** ECS disk snapshot → OSS export → download. Remove `aliyun-cli` agent, fix fstab from `/dev/vda` (already virtio!) to UUID
- **OpenStack:** Glance image download or Cinder volume snapshot export. Disable cloud-init OpenStack datasource, fix fstab, minimal initramfs changes (often already QCOW2/VirtIO)
- **Common:** All cloud exports get fstab UUID repair, initramfs rebuild, GRUB fix, network cleanup, serial console enable

h2kvm auto-detects the source cloud based on installed agents, driver stack, and fstab patterns. It applies the correct set of fixes for each source — no manual configuration needed. Just point it at the exported disk image.

Repatriation Scenarios

Real-world cloud exit projects powered by h2kvm + hypersdk.

Finance — AWS Exit, 40 VMs

- **Source:** AWS EC2 (m5/r5 instances), RDS → self-managed PostgreSQL
- **Workloads:** Trading platform, risk analytics, PostgreSQL, Redis
- **Motivation:** \$1.2M/yr cloud bill, sub-ms latency needs, SOX audit complexity
- **Target:** On-prem KVM cluster (Dell PowerEdge + NVMe)
- **Timeline:** 4 weeks (10 VMs/week, off-hours cutover)
- **Result:** **\$960K/yr saved**, latency dropped from 12ms to <1ms

Healthcare — Azure Exit, 25 VMs

- **Source:** Azure D-series VMs, managed disks
- **Workloads:** EMR system, PACS imaging, HL7 integration engine
- **Motivation:** HIPAA data residency, \$480K/yr Azure spend, egress on imaging
- **Target:** On-prem KVM in hospital data center
- **Timeline:** 3 weeks
- **Result:** **\$420K/yr saved**, patient data physically on-site, HIPAA simplified

SaaS — Multi-Cloud to Single Cluster, 80 VMs

- **Source:** 35 on AWS, 25 on Azure, 20 on GCP
- **Workloads:** Microservices (Docker), PostgreSQL, Elasticsearch, Redis, Kafka
- **Motivation:** \$2.1M/yr combined cloud spend, 3x ops complexity, 3x security audits
- **Target:** On-prem K3s + KubeVirt (containers + VMs on one platform)
- **Timeline:** 6 weeks
- **Result:** **\$1.8M/yr saved**, one platform to manage, one security framework

Government — GCP Exit, 15 VMs (Classified)

- **Source:** GCP Compute Engine (commercial cloud)
- **Workloads:** Data processing, analytics, internal apps
- **Motivation:** New classification requirements — data must move to air-gapped facility
- **Target:** Air-gapped KVM cluster (no internet, ever)
- **Timeline:** 2 weeks
- **Result:** **\$200K/yr saved**, full FedRAMP/STIG compliance, zero internet dependency

Retail — AWS Repatriation to Edge, 30 VMs

- **Source:** AWS EC2 running POS backends for 30 store locations
- **Workloads:** Inventory management, POS processing, reporting
- **Motivation:** \$540K/yr AWS + latency kills checkout speed + outages during connectivity loss
- **Target:** K3s + KubeVirt on mini PC at each store
- **Timeline:** 4 weeks (convert at HQ, ship hardware to stores)

Manufacturing — OCI + Azure to On-Prem, 20 VMs

- **Source:** 12 on OCI (SCADA analytics), 8 on Azure (MES dashboards)
- **Workloads:** Time-series DBs, Grafana, quality control, MES integration
- **Motivation:** Factory floor needs sub-10ms, egress on sensor data, OCI contract expiring
- **Target:** Ruggedized edge server + KVM at each plant
- **Timeline:** 3 weeks

- **Result: \$500K/yr saved**, POS works offline, sub-ms transaction time

- **Result: \$320K/yr saved**, real-time factory analytics, no internet dependency

Why No Other Tool Can Do This

Cloud repatriation is the one migration direction that cloud vendors don't help with.

Capability	h2kvm + hypersdk	virt-v2v	MTV	AWS MGN	Azure Migrate	Manual
Export from AWS	✓	✗	✗	Inbound only	✗	Complex
Export from Azure	✓	✗	✗	✗	Inbound only	Complex
Export from GCP	✓	✗	✗	✗	✗	Complex
Export from OCI	✓	✗	✗	✗	✗	Complex
Multi-cloud in one tool	✓ (10 providers)	✗	✗	✗	✗	✗
Remove cloud agents	✓ Auto	✗	✗	N/A	N/A	Manual
Guest OS fixes	✓ Auto	✓	Limited	✗	✗	Manual
Deploy to KVM	✓	✓	✗	✗	✗	Manual
Deploy to KubeVirt	✓	✗	✓	✗	✗	Complex
Boot verification	✓	✗	✗	✗	✗	Manual
Batch / scheduling	✓	✗	✓	✓	✓	✗
Cost	\$0	\$0	\$50K+	Free/90d	Free tool	\$0

The Critical Insight

Cloud vendors build tools to migrate INTO their cloud, not OUT. AWS MGN migrates to AWS. Azure Migrate migrates to Azure. Neither helps you leave. virt-v2v doesn't support any cloud source. MTV only handles VMware.

h2kvm + hypersdk is **the only tool purpose-built for cloud repatriation** — exporting FROM clouds TO on-premises infrastructure. It's the tool cloud vendors don't want to exist.

10 cloud providers. Automated export + conversion + guest fixes + deployment.

\$600K-\$1.8M saved per project. One tool. Zero lock-in. Free forever.

h2kvm + hypersdk: the cloud exit platform.

h2kvm + hypersdk — Enterprise VM Migration Platform | Proprietary (Zyvor AI Labs) | github.com/zyvorai/h2kvm